Thibeau Wouters

Personal information

Full name:	Thibeau Wouters	Email address:	thibeauwouters@gmail.com
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Research interests

Gravitational waves & binary neutron star mergers — Neutron star equation of state
Machine learning & differentiable programming — GPU-accelerated Bayesian inference

Professional experience

2023–2027 **PhD Candidate in Physics** — Utrecht University
Advisor: Prof. dr. Chris Van Den Broeck

Education

2023 **M.Sc. in Artificial Intelligence** *(89.00% summa cum laude)* — KU Leuven
Advisors: Prof. dr. Tjonnie G. F. Li & Prof. dr. Johan Suykens

2022 **M.Sc. in Theoretical Physics (minor: nuclear physics)**
(91.41% summa cum laude with congratulations of the Board of Examiners) — KU Leuven
Advisor: Prof. dr. Nikolay Bobev

2020 **B.Sc. in Physics** *(74.42% cum laude)* — KU Leuven

2020 **B.Sc. in Mathematics** *(73.92% cum laude)* — KU Leuven

Publication and citation summary

	Excluding LVK/ET† papers	Including LVK/ET† papers
Number of papers:	**21**	61
Number of citations:	**224**	3,401
Average citations per paper:	**10.7**	55.8
h-index:	**10**	24

†*LVK = LIGO-Virgo-KAGRA Collaboration, ET = Einstein Telescope Collaboration*

Selected publications and preprints (links to full list, excl. collaboration papers)

F. Gittins, H. Narola, T. Wouters, et al. (2026). Accepted at Physical Review Letters. arXiv: `2606.06376 [gr-qc]`
— Projection of detectability of neutron star resonance modes with the Einstein Telescope

T. Wouters et al. (2026). ***Phys. Rev. D*** **114.4, p. 043005**
— Physics-informed priors constructed with machine learning for gravitational-wave inferences

H. Koehn, T. Wouters, et al. (2025). ***Astron. Astrophys.*** **704, A55**
— Bayesian inference framework (FIESTA) for efficient analysis of kilonovae and GRB afterglows

T. Wouters et al. (2025). ***Phys. Rev. D*** **112.4, p. 043037**
— Introduces JESTER, an autodifferentiable solver for the Tolman–Oppenheimer–Volkoff equations and GPU-accelerated inference of the neutron star equation of state

T. Wouters et al. (2024). ***Phys. Rev. D*** **110.8, p. 083033**
— GPU-accelerated inference of gravitational-wave signals from binary neutron star mergers (JIM)

Collaboration work

— Lead developer of JESTER, reviewed within the LIGO-Virgo-KAGRA Collaboration
— Analyst of the 6th gravitational-wave transient catalog of the LIGO-Virgo-KAGRA Collaboration
— Contributed to geometry comparison report for the Einstein Telescope Collaboration, to be submitted to the board of governmental representatives, using JESTER
— Leading a task force in the Einstein Telescope Nuclear Physics Division on equation-of-state inference systematics, using JESTER

Awards and prizes

2024–2026 SURF small computing grants (3 million CPU-core computing hours total)
2023 KU Leuven Short Mobility Grant (EUR 500)
2023 Belgian Physical Society best master thesis in physics (EUR 100)
2022 KU Leuven Physics prize: highest grade among physics master students

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Talks

Invited talks and workshops

2027	Einstein Telescope Early Career Workshop *(upcoming)*	Amsterdam, Netherlands
2027	Differentiable Programming: JAX for Science *(upcoming)*	Leiden, Netherlands
2027	Constraining the Nuclear Matter Equation of State *(upcoming)*	Mainz, Germany
2025	Research Seminar	AEI Potsdam, Germany
2025	Accelerated Computing for Application Workflows	SURF, Netherlands
2025	Advanced Computing User Day	SURF, Netherlands
2024	JAXtronomy: JAX for astronomy research	New York, USA

Conference talks

2026	Belgian-Dutch GW meeting (plenary talk)	Louvain-la-Neuve, Belgium
2026	GWFreeride	Sexten, Italy
2026	LIGO-Virgo-KAGRA collaboration meeting	Pisa, Italy
2025	European AI for Fundamental Physics Conference	Sardinia, Italy
2026	Belgian-Dutch GW meeting	Nijmegen, Netherlands
2025	3rd AIslands workshop on AI in gravitational waves	Bute, Scotland
2025	XIV Einstein Telescope Symposium	Bologna, Italy
2025	Dutch Physics Society	Lunteren, Netherlands
2024	Extreme Matter in Extreme Stars	Leiden, Netherlands
2024	2nd AIslands workshop on AI in gravitational waves	Arran, Scotland
2024	Nikhef Annual Scientific Meeting	Amsterdam, Netherlands
2023	Machine learning for gravitational waves workshop	Utrecht, Netherlands

Teaching record

Student supervision

Master

2026	Frank Snijder
2025	Marlinde Drent
2025	Oleksii Bevza
2025	Hong Ye Guo

Bachelor

2026	Elin Jakobs
2025	Senna van Os
2024	Frank Snijder
2024	Tom Oeloff
2024	Marlinde Drent

Courses tutored

2025	Master	Machine learning	Utrecht University
2024	Bachelor	Gravitational waves	Utrecht University
2024	Master	Gravitational waves	Utrecht University
2023	Bachelor	Optics and astronomy	Utrecht University
2021	Master	Quantum mechanics	KU Leuven
2020–2022	Bachelor	Introductory mathematics and physics	KU Leuven

Referee work

- Physical Review D
- Journal of Open Source Software

Public and professional services

2026 Student representative in the Faculty Open Science Team, Utrecht University
2025 Volunteer Einstein Telescope stand at Nerdland, the largest Belgian popular science festival
2024 Outreach lecture on space-time for students of HKU University of Arts, Utrecht
2023 Student representative responsible for the quality of education of the AI master, KU Leuven

References

- Prof. dr. Chris Van Den Broeck, Utrecht University, c.f.f.vandenbroeck@uu.nl
- Prof. dr. Tim Dietrich, University of Potsdam, tim.dietrich@uni-potsdam.de
- Prof. dr. Michael Coughlin, University of Minnesota, cough052@umn.edu